

Linked List (Assignment Solutions)

Solution 1:

```
int getSize(ListNode* head) {  
    int size=0;  
    while(head != NULL) {  
        head = head->next;  
        size++;  
    }  
  
    return size;  
}  
  
ListNode *getIntersectionNode(ListNode *headA, ListNode *headB) {  
    int m = getSize(headA);  
    int n = getSize(headB);  
    ListNode* t1 = headA;  
    ListNode* t2 = headB;  
    int diff = 0;  
  
    if(m >= n) {  
        diff = m-n;  
        for(int i=0; i<diff; i++) {  
            t1 = t1->next;  
        }  
    } else {  
        diff = n-m;  
        for(int i=0; i<diff; i++) {  
            t2 = t2->next;  
        }  
    }  
  
    while(t1 != NULL && t2 != NULL && t1 != t2) {
```

```
        t1 = t1->next;
        t2 = t2->next;
    }

    if(t1 == NULL) {
        return NULL;
    } else {
        return t1;
    }
}
```

Solution 2:

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```
ListNode* deleteNodes(ListNode* head, int m, int n) {
    auto pre = head;
    while (pre) {
        for (int i = 0; i < m - 1 && pre; ++i) {
            pre = pre->next;
        }
        if (!pre) {
            return head;
        }
        auto cur = pre;
        for (int i = 0; i < n && cur; ++i) {
            cur = cur->next;
        }
        pre->next = cur ? cur->next : nullptr;
        pre = pre->next;
    }
    return head;
}
```

Solution 3 :

```
int getSize(ListNode* head) {
    int size = 0;
    while(head != NULL) {
        size++;
        head = head->next;
    }

    return size;
}

ListNode* KthFromEnd(ListNode* head, int k) {
    ListNode* temp = head;
    int size = getSize(head);
    for(int i=0; i<size-k; i++) {
        temp = temp->next;
    }

    return temp;
}

ListNode* swapNodes(ListNode* head, int k) {
    ListNode* left = head;
    for(int i=0; i<k-1; i++) {
        left = left->next;
    }

    ListNode* right = KthFromEnd(head, k);

    int temp = left->val;
    left->val = right->val;
    right->val = temp;
    return head;
}
```

Solution 4 :

```
ListNode* oddEvenList(ListNode* head) {
    //case of LL length = 0, 1, 2
    if(head == NULL || head->next == NULL || head->next->next == NULL) {
        return head;
    }

    ListNode *evenStart = head->next;
    ListNode *odd = head;
    ListNode *even = head->next;

    while(odd->next && even->next){
        odd->next = even->next;
        even->next = odd->next->next;
        odd = odd->next;
        even = even->next;
    }
    odd->next = evenStart;
    return head;
}
```

Solution 5 :

```
ListNode* merge(ListNode* list1, ListNode* list2) {
    ListNode* merged = new ListNode(-1);
    ListNode* mptr = merged;
    ListNode* left = list1;
    ListNode* right = list2;

    while(left != NULL && right != NULL){
        if(left->val <= right->val){
            mptr->next = left;
            mptr = mptr->next;
        }
    }
}
```

```

        left = left->next;
    }
    else{
        mptr->next = right;
        mptr = mptr->next;
        right = right->next;
    }
}

if(left != NULL){
    mptr->next = left;
}

if(right != NULL){
    mptr->next = right;
}

return merged->next;
}

ListNode* mergeKLists(vector<ListNode*>& lists) {
    if(lists.size() == 0){
        return NULL;
    }
    if(lists.size() == 1){
        return lists[0];
    }
    ListNode* head = lists[0];
    for(int i=1; i<lists.size(); i++){
        head = merge(head, lists[i]);
    }

    return head;
}

```